

EXAM 7 – SPRING 2011

4. (2 points)

Given the following loss development triangle:

<u>Year</u>	<u>Cumulative Reported Losses (\$000)</u>									
	<u>Age in Years</u>									
	<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>
0	5,400	10,800	17,280	24,192	29,030	33,385	36,056	37,498	37,873	37,873
1	5,800	12,039	19,898	28,773	37,369	46,089	54,105	58,895	64,579	
2	3,888	7,582	12,199	16,834	20,933	24,367	27,335	29,207		
3	4,900	9,930	16,062	22,669	27,533	32,092	35,391			
4	5,655	10,631	17,123	23,802	29,320	33,980				
5	4,950	10,131	16,568	23,712	29,215					
6	6,650	13,475	22,073	30,902						
7	4,890	10,269	16,512							
8	5,780	11,502								
9	4,964									

A linear regression of the incremental losses against the previous cumulative losses was performed yielding the following table:

	<u>0 to 1</u>	<u>1 to 2</u>	<u>2 to 3</u>	<u>3 to 4</u>	<u>4 to 5</u>	<u>5 to 6</u>
Constant a	-66	-733	-740	-2,638	-4,183	-4,247
Standard Dev. of a	920	442	960	2,494	2,305	3,195
Factor b	1.024	0.692	0.453	0.352	0.323	0.250
Standard Dev. of b	0.171	0.041	0.055	0.106	0.079	0.092

a. (1 point)

Use the data above to demonstrate whether the chain ladder method or the linear plus a constant method better models the emergence of losses.

b. (1 point)

Support graphically the answer from part a. above.

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Sample Solutions
Exam 07, 2011, Question 4

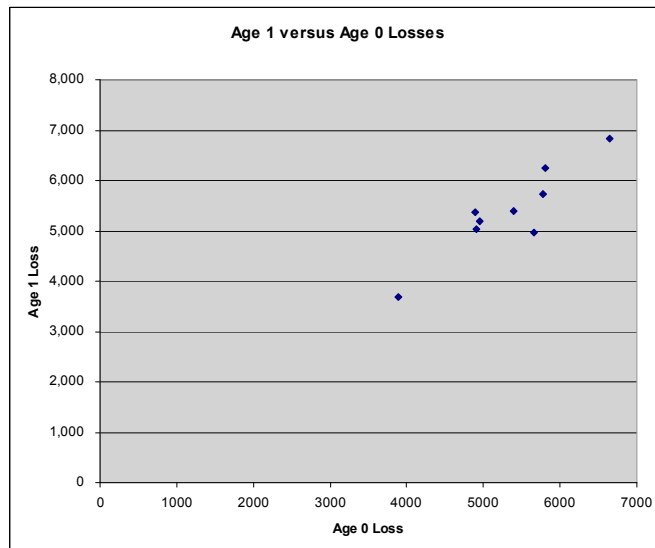
Solution 1

a.)

Significance of Factors

Usually the absolute value of a factor is required to be at least twice its standard deviation for the factor to be significant. So the table suggests that the b factors for the chain ladder method are significant. The chain ladder model is thus supported by this data.

- b.) One can plot Age d+1 incremental losses against Age d cumulatives. The chain ladder results should be something close to a straight line through the origin with slope equal to the development factor.



A factor by itself would be a straight line through the origin with slope equal to development factor. The above chart suggests the chain ladder is a better model.